THESIS TITLE

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DEPARTMENT

Submitted to University College London (UCL) in partial fulfilment of the requirements for the degree of

Doctor of Philosophy

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WORD COUNT: 904

Declaration

I, AUTHOR FULL NAME, confirm that the work presented in this thesis is my own. Where information has been derived from other sources, I confirm that this has been indicated in the thesis.

Signature:

Date:

Abstract

The signed declaration should be followed by an abstract consisting of no more than 300 words. The abstract can be edited inside `main.typ`.

Impact Statement

The abstract should be followed by an impact statement consisting of no more than 500 words. For further information on the content of the Impact Statement, please see the Impact Statement Guidance Notes for Research Students and Supervisors. See https://www.ucl.ac.uk/study/sites/study/files/impact-statement-guidance-notes.pdf. The impact statement can be edited inside `main.typ`.

Acknowledgements

The acknowledgements can be edited inside `main.typ`.

The dedication can be edited inside `main.typ`*.*

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Table of Symbols

Notation	Description
T	Temperature in Kelvin

Table of Acronyms

Notation	Description
T	Temperature in Kelvin

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Section 1
DEMO

This section is meant to detail the features in this thesis template and necessary steps you should take. The thesis is inspired by "Pawel's LaTeX template". Like their template, this PhD thesis is geared towards PhDs in STEM fields.

1. This section is inside `main.typ` as `#include "demo.typ"`. This line should be deleted before submission. You can optionally delete this file afterwards or turn it into something useful.
2. After reading this document, please take some time to read through the comments littered around in `main.typ`, the project root file, and `ucl-thesis.typ` which contains the style code that makes up this thesis template. There are also a few convenience commands to mimic their LaTeX equivalent.
3. Please ensure to insert an image of your signature in the right section.
4. If you are editing locally, I recommend Visual Studio Code with the Tinymist extension (or any editor with support for it) as well as typstyle linter to format your Typst code. Even if you have a potato device, Typst compiles *really* quickly even for hundred+ page documents with many packages, which your thesis likely will be.
5. In this thesis, Typst's native `bibliography` is used, with an example bib file in `main.bib` - see also `main.typ` for how it's imported. The bib format is **biblatex**. An example citation is [1]
   - Instead of using default `american-physics-society` citation style, I have hacked it (thanks to the Typst wizards of the internet!) so that DOI/links are clickable (as per UCL guidelines) and coloured for visibility. Some things may be broken - please let me know or submit an issue on the repository.
   - Typst by default doesn't have support for bibliographies across multiple files. If you need multiple bibliographies in general, the

alexandria package may be useful and you can import it in `ucl-thesis.typ`. However, I have used another hack to get the same bibliography: simply call `#load-bib()` at the end of each subfile. If you change from the default `main.bib`, call it as `#load-bib(bib-file: "file.bib")`.

6. To include more `.typ` files, see files in various directories and also `main.typ` on how to use them.
7. The total wordcount is automatically handled by the wordometer package. As per the RDF: "The maximum word lengths include footnotes, tables and figures but exclude bibliography, appendices and supporting data" and the template has excluded the appendices and bibliography.

1.1 Features

This sub-section aims to demonstrate the default packages I have included.

Your thesis is a living document. It will contain your research over the years. It's easy to make a note on your phone, a different software, pen and paper etc but it's also hard to find that same information much later down the line, so why not note in the document! This is achieved by the drafting package.

Cite this sentence

As a thesis targeted towards STEM, there are a lot of STEM-related packages. As a computational physicist myself, I write a lot of math - the physica package makes writing derivatives and other stuff a lot more convenient:

$$\begin{aligned}\hat{H} = &-\frac{\hbar^2}{2m_e}\sum_i \nabla_i^2 - \sum_{i,I}\frac{Z_I e^2}{|\boldsymbol{r}_i - \boldsymbol{R}_I|} + \frac{1}{2}\sum_{i\neq j}\frac{e^2}{|\boldsymbol{r}_i - \boldsymbol{r}_j|} \\ &-\sum_i \frac{\hbar^2}{2M_I}\nabla_I^2 + \frac{1}{2}\sum_{I\neq J}\frac{Z_I Z_J e^2}{|\boldsymbol{R}_I - \boldsymbol{R}_J|}\end{aligned} \tag{1}$$

which is written:

```
$
    hat(H) = & -frac(hbar^2, 2m_e)sum_i grad^2_i - sum_(i,I)
    frac(Z_I e^2, abs(bold(r)_i - bold(R)_I)) + frac(1, 2)
    sum_(i != j) frac(e^2, abs(bold(r)_i - bold(r)_j)) \ & -
```

```
    sum_i frac(hbar^2, 2M_I)grad^2_I + frac(1, 2) sum_(I !=
    J) frac(Z_I Z_J e^2, abs(bold(R)_I - bold(R)_J))
$ <eqn:fullschrodinger>
```

The `grad` command is part of `physica`, and this code block was beautifully displayed by codly!

quill provides the ability to draw quantum circuits!

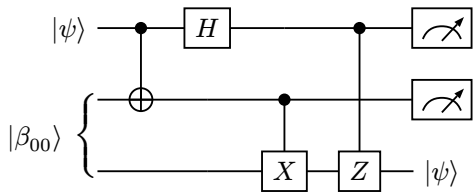

Figure 1: Quantum teleportation circuit from the Quill user manual.

If you scroll up to the list of figures, you'll see this has automatically been included! Any figure considered an image (e.g. most things. Tables go in a separate section) will be automatically added! The same for tables, and the same for algorithms

Table 3: Unit conversion for common quantities in atomic physics.

Quantity	Atomic unit	SI unit	Conversion
Energy	Hartree, Ha	Electronvolt, eV	$1\text{Ha} = 27.2114079527$ eV
Distance	Bohr, a_0	Å	$1a_0 = 0.529177249$Å

Algorithm 1: Steepest Descent

```
procedure STEEPEST DESCENT(R_0, ε)
    ▷ Requires: f differentiable, initial point R_0 ∈ ℝ^n, tolerance ε > 0
    k ← 0

    while ‖∇f(x_k)‖ > ε do
        x_{k+1} ← x_k − α_k ∇f(x_k)
        k ← k + 1
    end
    return x_k
end
```

And you can of course refer to Algorithm 1 or Table 3.

Finally, here's the UCL logo as an example figure

Figure 2: The UCL logo from the title page at 80% width.

Chapter 2
Introduction

2.1 Background and Motivation

2.2 Research Questions

2.3 Thesis Outline and Contributions

Chapter 3
Theory

Chapter 4
Experiment

Chapter 5
Conclusions

5.1 Summary of Contributions

5.2 Limitations

5.3 Future Directions

Appendix A
This has a title

This can be for extra information, figures, publications that was not directly relevant to your thesis.

Appendix B

This Appendix B refers to Appendix A.

Here is an equation:

$$x = \frac{-b}{2a} \pm \frac{\sqrt{b^2 - 4ac}}{2a} \tag{2}$$

It is Equation 2

Bibliography

[1] G. Alvarez, 'Quantum Mechanics as Classical Mechanics plus Quantum Corrections: The Cubic Anharmonic Oscillator', *Journal of Physics A: Mathematical and General*, vol. 22, no. 6, pp. 617–624, Mar. 1989, doi: 10.1088/0305-4470/22/6/014.